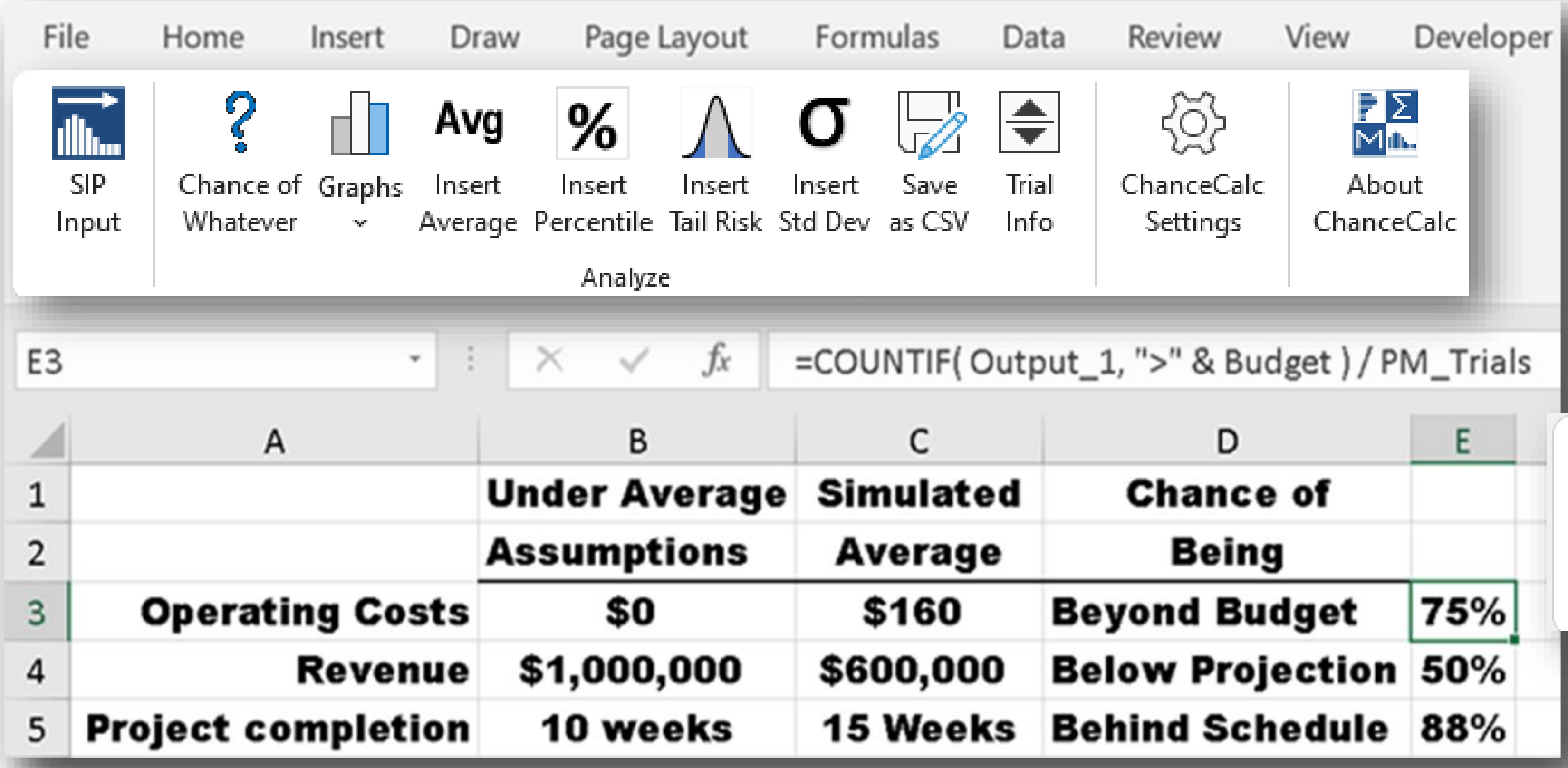
Probability Management

2/15/2023

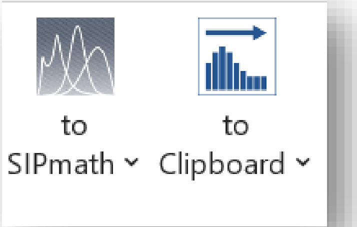
Getting Started With Chance™ Calc +

[**Click Here For Contents**]

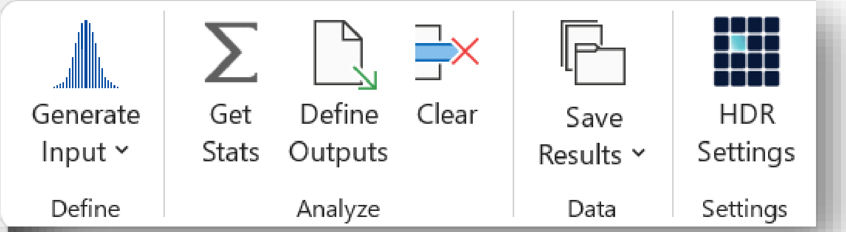
Note:
If you get a “Compile error in hidden module”, you may need to enable ActiveX controls.
See Appendix



Metalog



Monte Carlo



Probability Management

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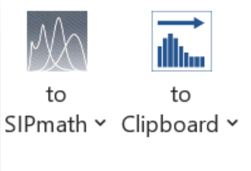

a) Select mode
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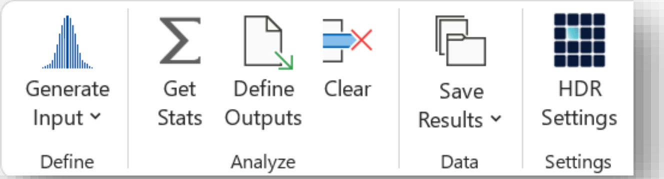

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1. Introduction

ChanceCalc is an Excel add-in for creating chance-informed dashboards in your spreadsheet based on uncertainties stored in **Stochastic Libraries** as CSV files, or in the open SIPmath™ 2.0 or 3.0 Standard. The models created with ChanceCalc are stand-alone Excel files that perform hundreds or thousands of simulation trials per key-stroke and do not require the add-in to run. It include a separate tab for importing Metalog distributions for Tom Keelin's Metalog Templates.

ChanceCalc Monte Carlo expands the capabilities to include Monte Carlo simulation with internal generation of random variates and the ability to save stochastic libraries as SIPmath 2.0 files, including metadata.

The SIPmath™ 3.0 Standard is based on Tom Keelin's revolutionary **Metalog distribution** and is driven by Doug Hubbard's **HDR portable random number generator**. Streams of up to 100 million trials of a very wide variety of probability distribution may be encoded into 18 Metalog parameters and 4 HDR seeds. The open SIPmath 3.0 Standard wraps these 22 numbers in a JSON object, which can generate identical streams of random variates in any computer environment.

The SIPmath 3.0 Standard

The Metalog Distribution

The HDR Generator

2. ChanceCalc Ribbon

a. Basic Commands

Determine the chance that any formula is greater or less than any specified target

Determine the simulated average of any formula

Determine the simulated average of results beyond a specified target

Trial Info

Settings, including number of trials & Chance percent rounding

Select SIP Libraries and input into your model

Create histograms or cumulative charts of any formula or scatter plots of any pair of formulas

Determine a specified percentile of any formula

Standard Deviation

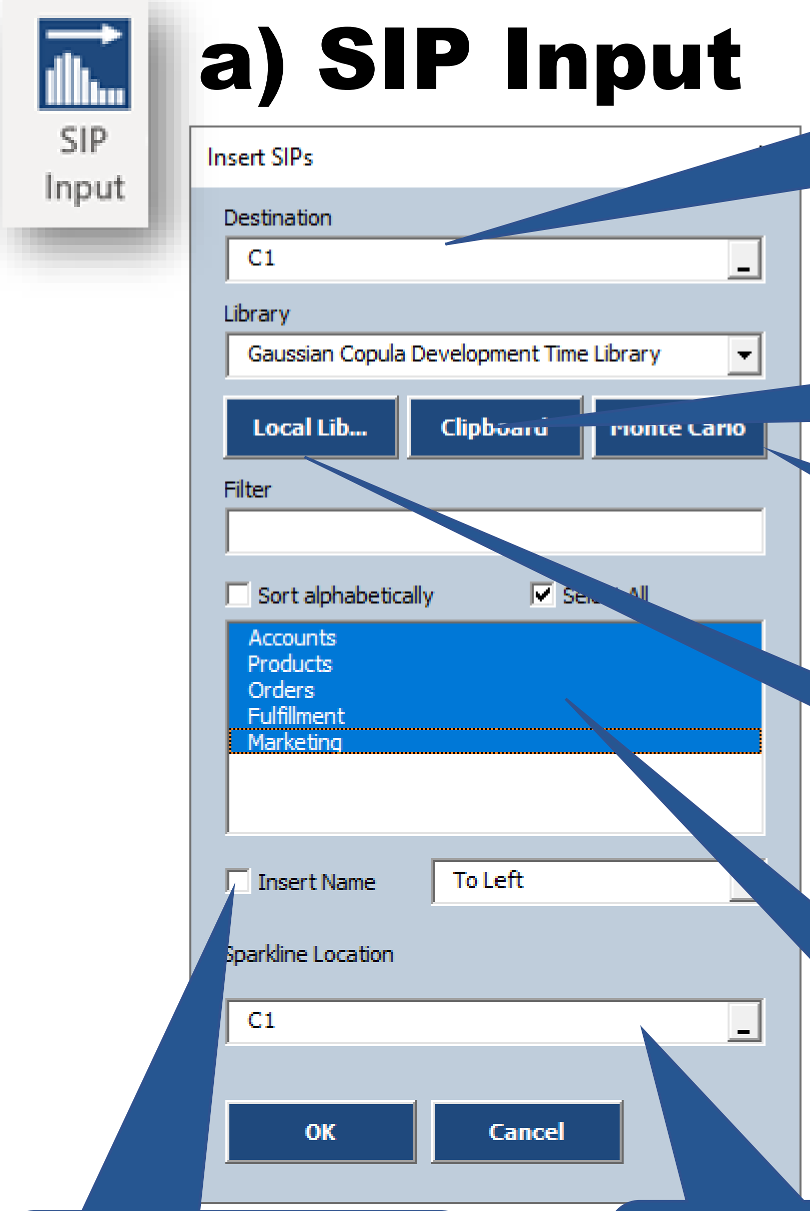
Save results as CSV file

Check version number

a) SIP Input

1. Select destination cells. It saves time to do this before clicking SIP Input

Libraries types supported are CSV, SIPmath™ 2.0 and 3.0 Standard and may be accessed from the web or your computer.

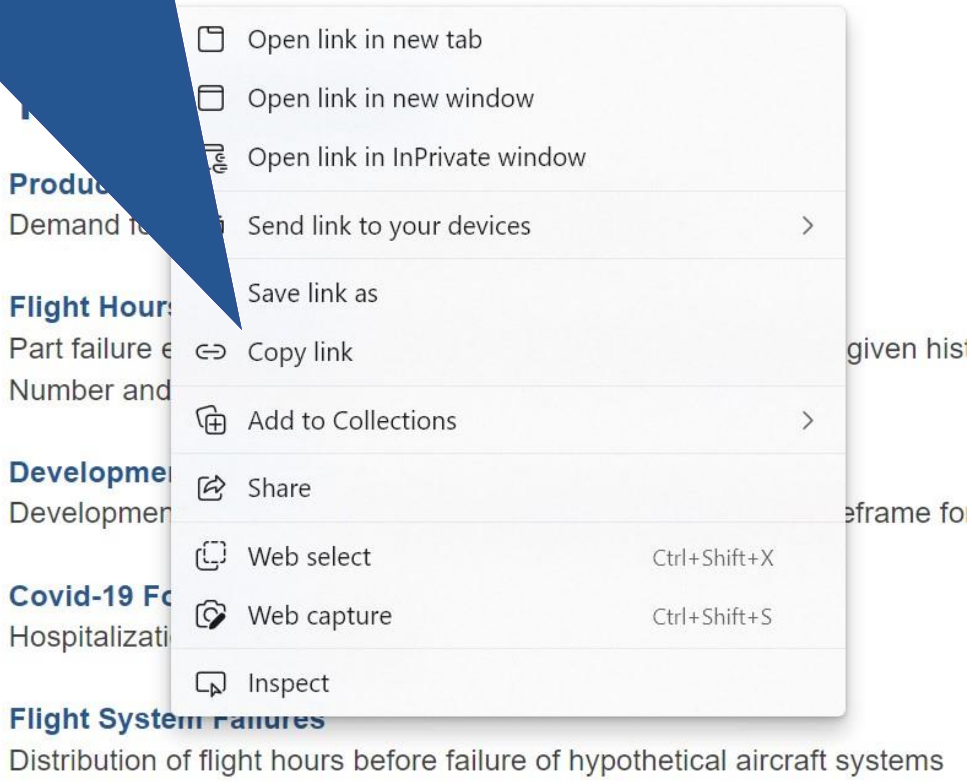


2a. Access a copied SIP library after right clicking and copying Link from online library

2c. Create SIPs with Excel formulas (CC Monte Carlo only)

2b. Open a SIP library from your local machine

3. Select SIPs to Insert



Optionally insert SIP names

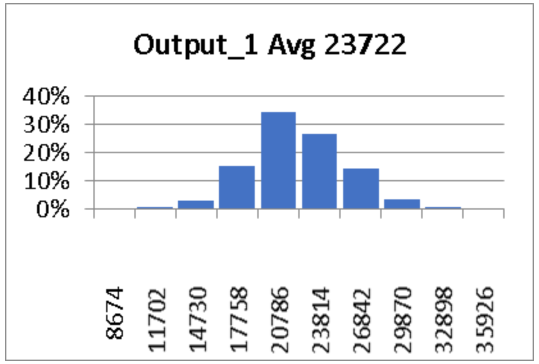
Optionally specify Sparkline location

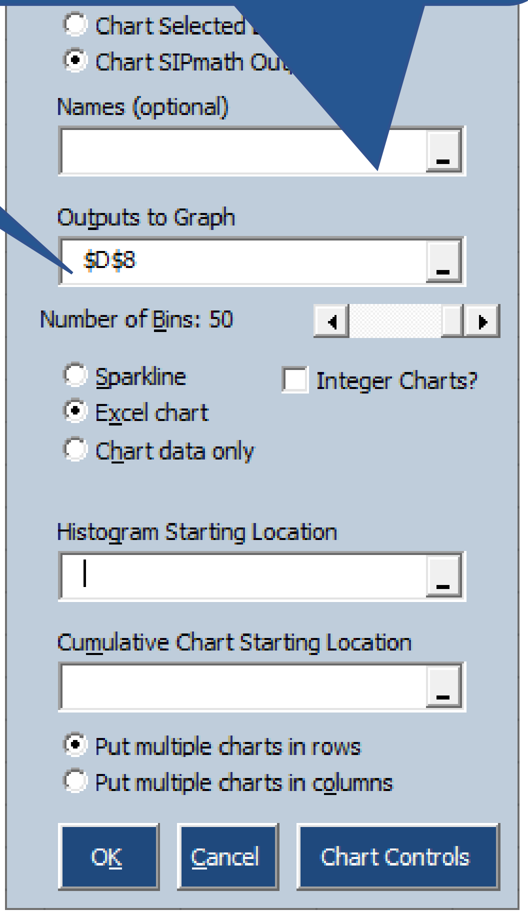
b) Graphs

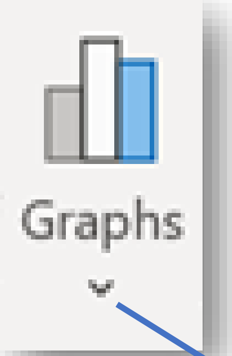
1. Select formula cell to graph before clicking button

2. Click button to create histograms, cumulative graphs

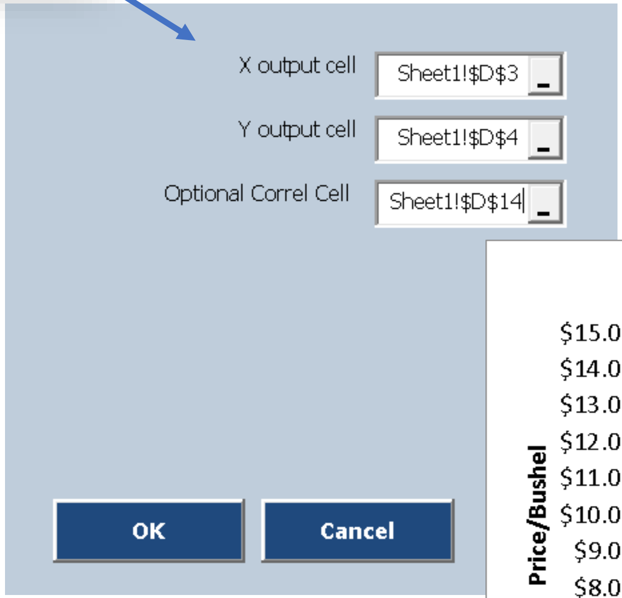
3. You may optionally name formulas when you graph them. This is useful if you plan to save your outputs.

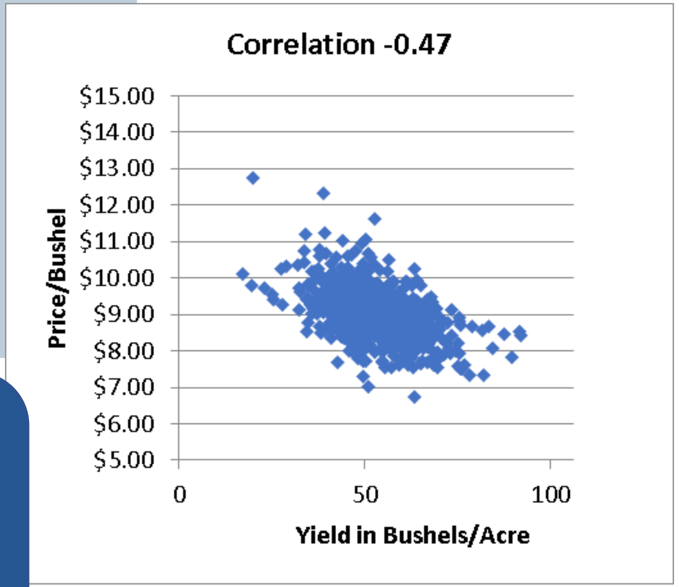






4. Click down arrow for scatter plots





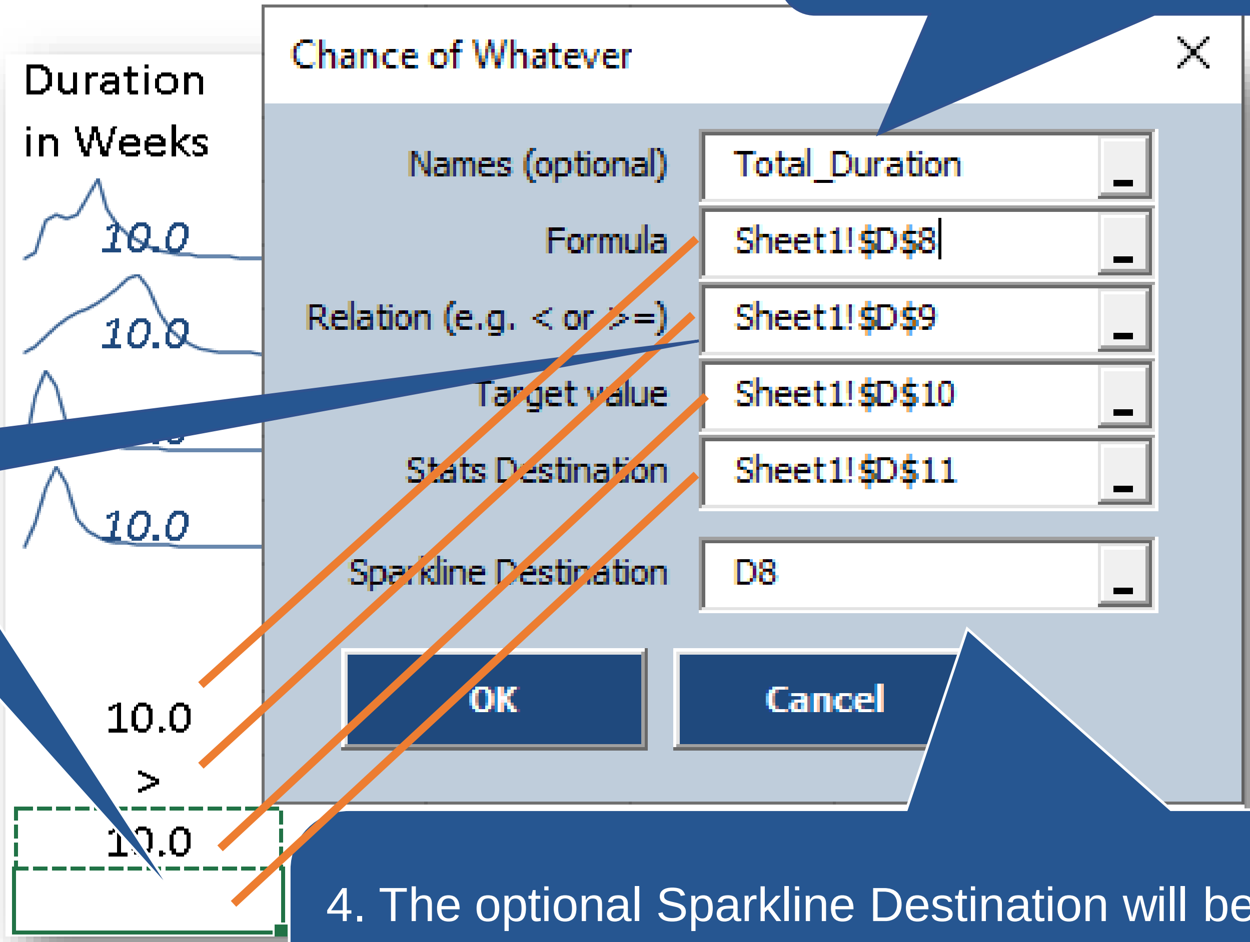
NOTE: the model will be automatically initialized, and outputs automatically defined, if necessary to make a graph.

c) Chance of Whatever

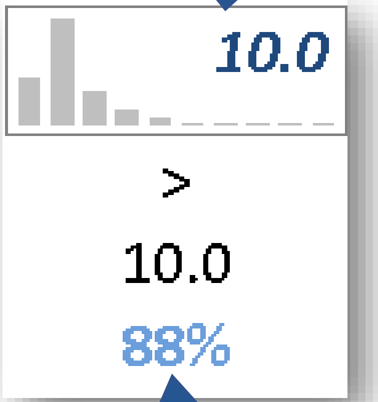
3. You may optionally name formulas when you get statistics on them. This is useful if you plan to save your outputs.

1. Select location of Chance cell before clicking button

2. Fill in fields and click **OK**



Formula will be simulated with a resulting sparkline histogram. Text will be *Italics* to indicate a single trial of many.



NOTE: ALL Stats button on this ribbon will automatically initialize the model, and automatically define formula cells as outputs, if needed to make a graph.

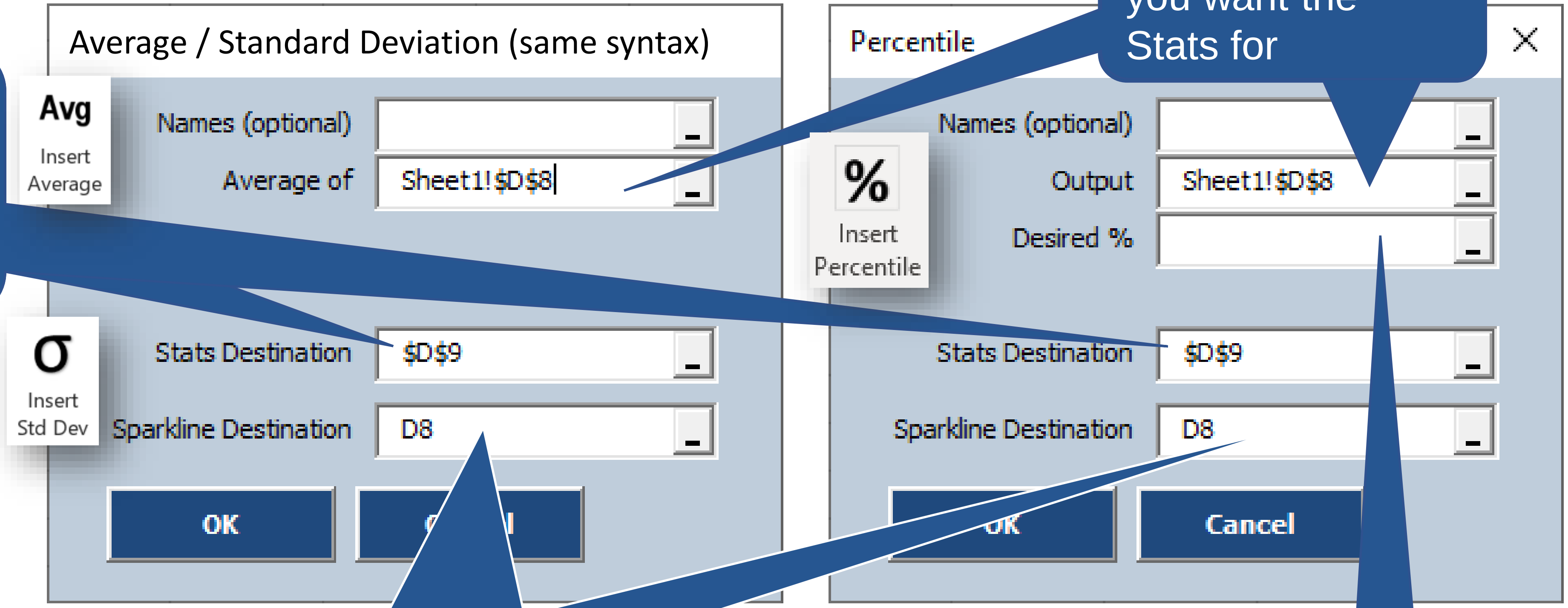
4. The optional Sparkline Destination will be auto-filled based on the Formula address and the Default Sparkline Position in the Settings dialog. You may also manually adjust it.

Chance appears in Blue to indicate a statistic calculated over all trials.

c) contd. Insert Avg, Standard Deviation, & Percentile

1. Select location of Stats before clicking button

2. Select Formula you want the Stats for



Reminder: You may optionally name formulas when you get statistics on them. This is useful if you plan to save your outputs.

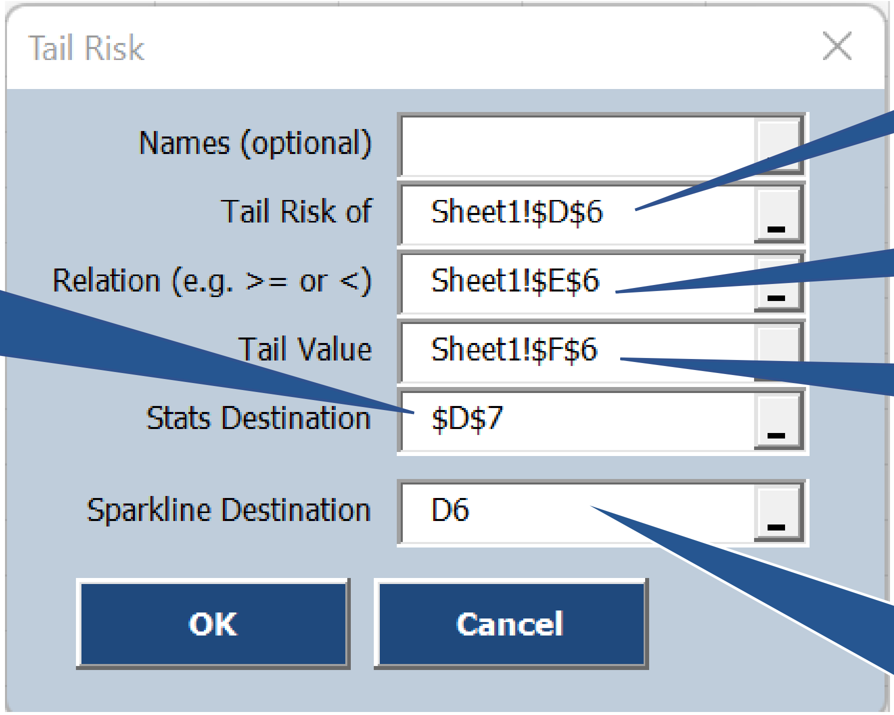
4. The optional Sparkline Destination will be auto-filled based on the Formula address and the Default Sparkline Position in the Settings Dialog. You may also manually adjust it.

Stats appears in blue to indicate a statistic calculated over all trials.

3. Select desired percent

c) contd. Insert Tail Risk

(The Average Above or Below a Specified Value)



1. Select location of Tail Risk result before clicking button

2. Select Formula you want the Tail Risk for

3. Relation may be typed directly or refer to a cell or range

4. Desired tail value may be typed directly or refer to a cell or range.

5. The optional Sparkline Destination will be auto-filled based on the Formula address and the Default Sparkline Position in the Settings Dialog. You may also manually adjust it.

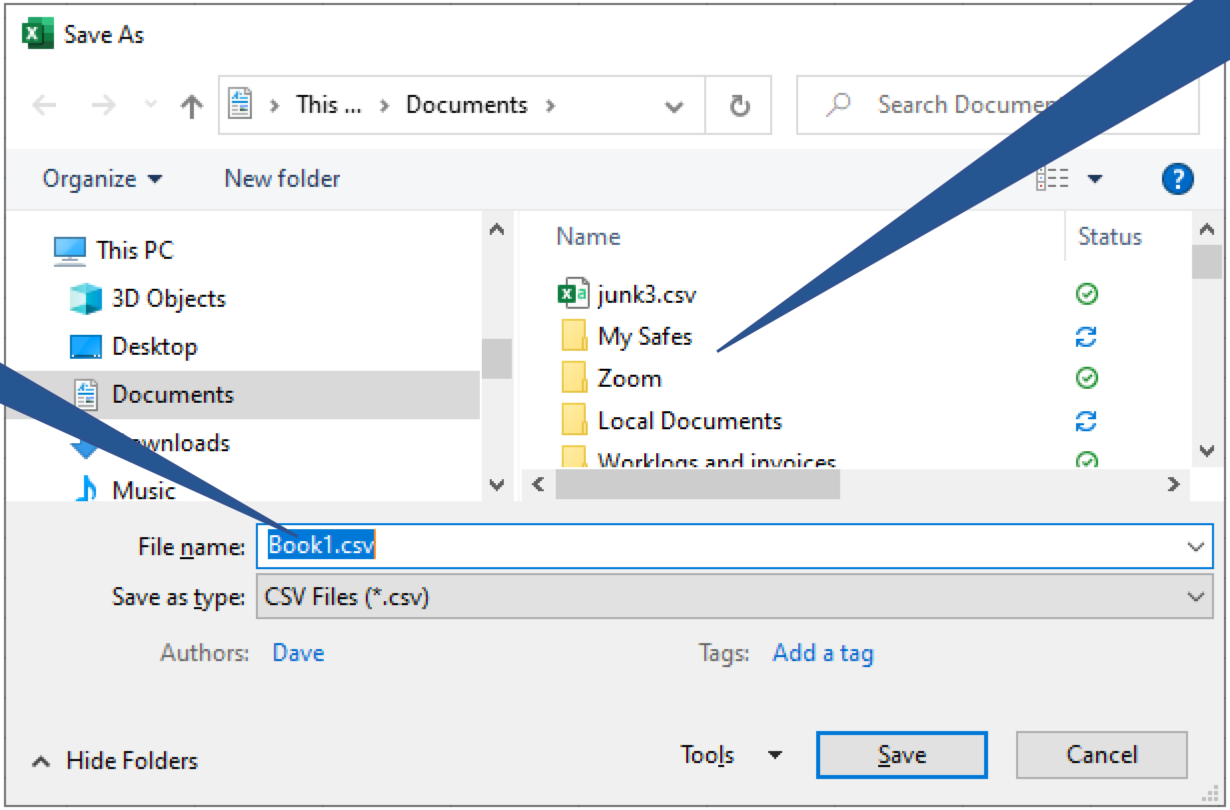
Reminder: You may optionally name formulas when you get statistics on them. This is useful if you plan to save your outputs.

Stats appears in blue to indicate a statistic calculated over all trials.

d) Save as CSV

1. Enter file name under which to save data

2. Select folder to save data

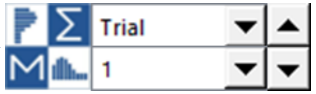


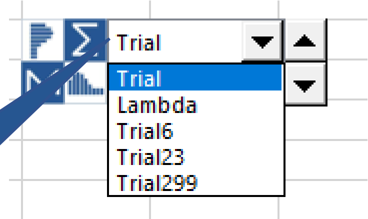
Simulated Data from the PMTable Tab will be saved with the name you specify

e) Trial Info

NOTE: You may specify the control to be automatically inserted upon model initialization from the Settings dialog.

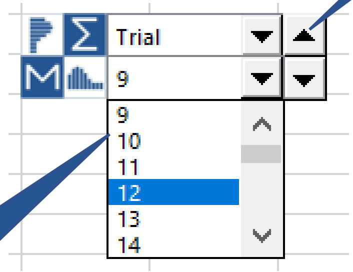
‘Floating’ controls that will be placed when pressing Trial Info button.



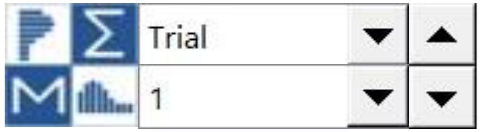


First drop-down selects display of either Trial numbers or Metadata values.

Spin button also changes trial number.



Second drop-down selects Trial number to display.

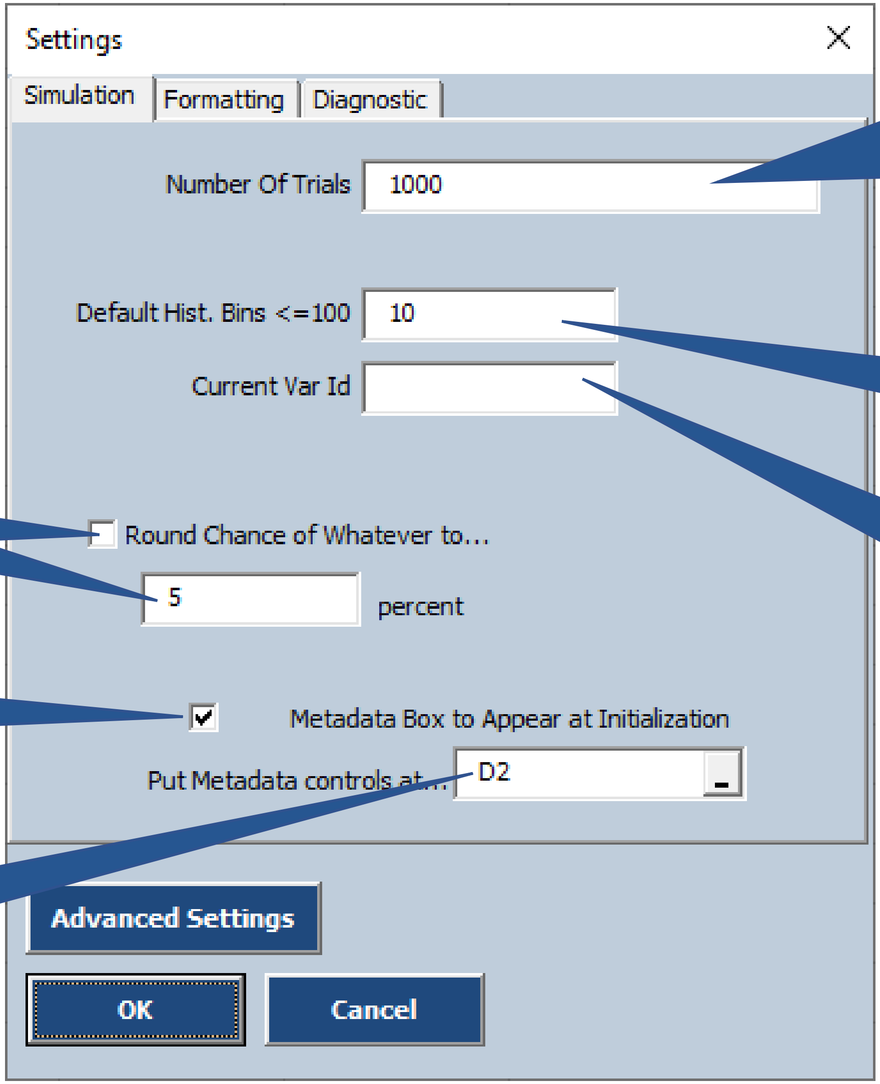


Trial Info control can be dragged to any desired position by clicking on the Logo.

f) Settings - Simulation

For Formatting and Diagnostic Settings

Click Here



Display number of trials (3.0 Libraries only). 2.0 Libraries are pre-specified.

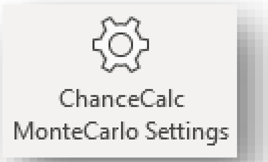
Default number of bins in a histogram.

Default Var ID value for HDR random numbers

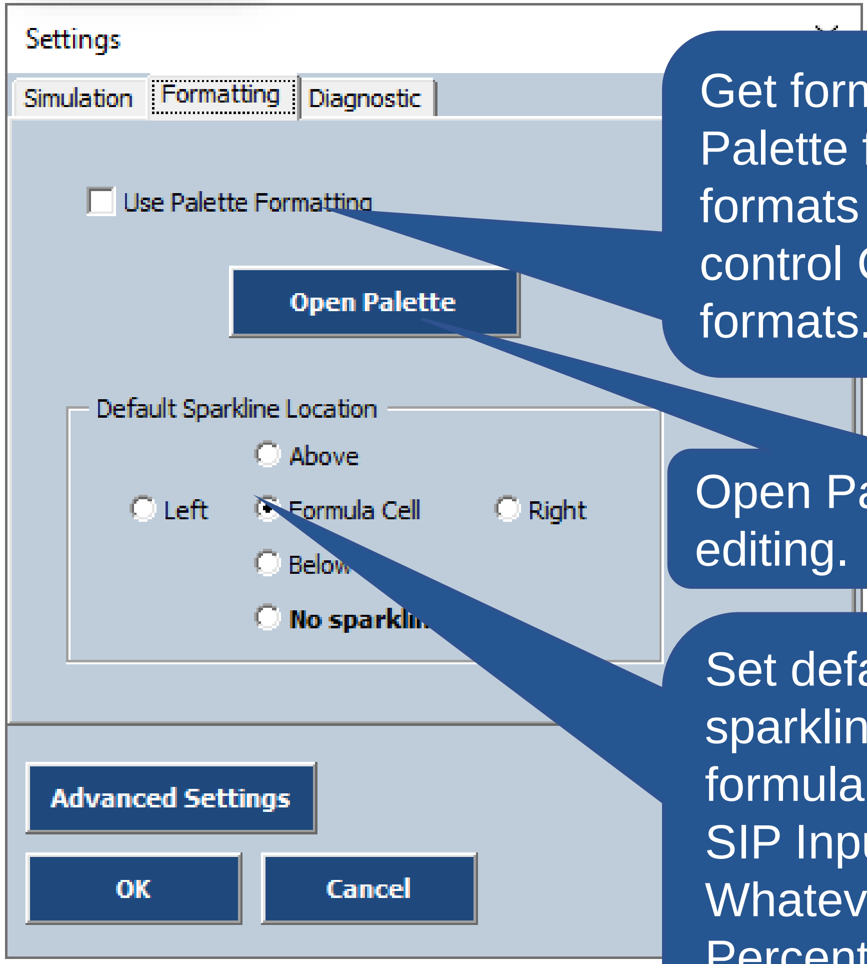
Limit precision on Chance of Whatever to avoid over specificity.

Check to automatically insert Metadata Trial Info Controls box.

Default location for Metadata Trial Info Controls box.



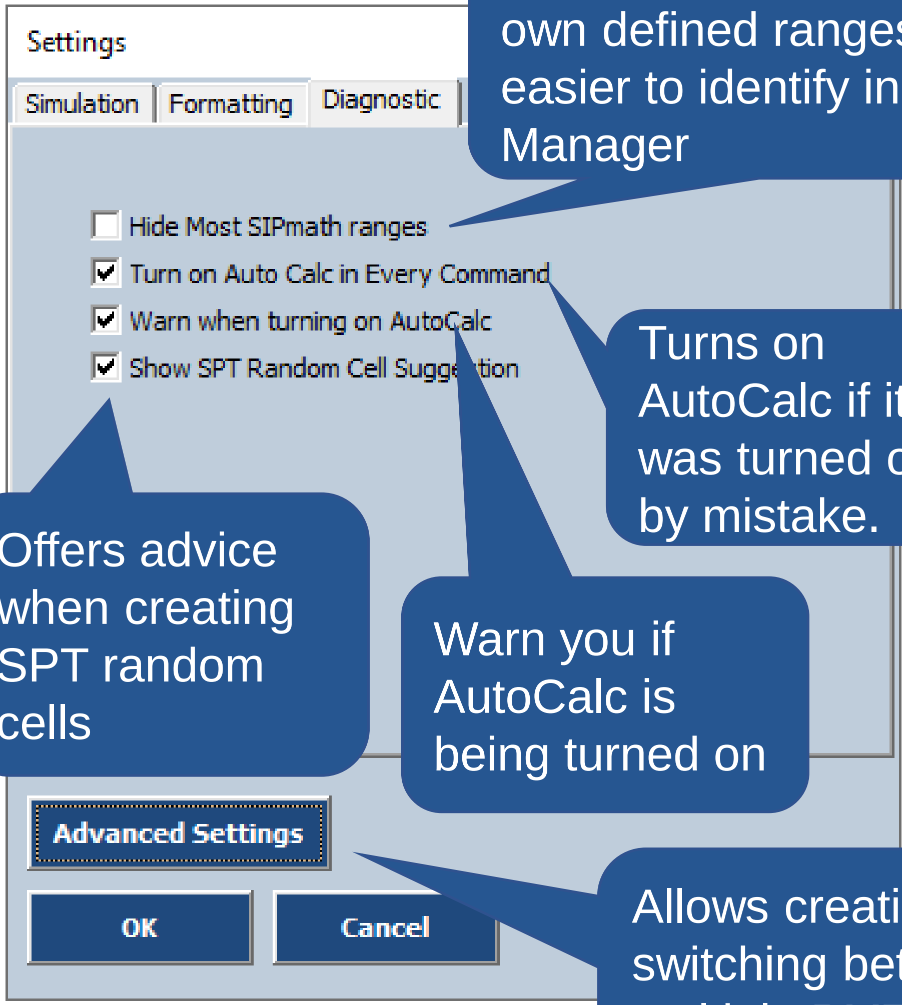
f) Settings – Formatting & Diagnostic



Get formatting from the Palette file. Change formats in the Palette to control ChanceCalc formats.

Open Palette file for editing.

Set default position of sparklines as relative to formula cells used in SIP Input, Chance of Whatever, Avg, Percentile, and Tail Risk.



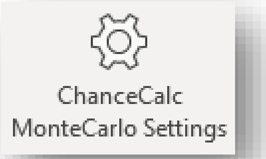
Hides internal names so your own defined ranges are easier to identify in the Name Manager

Turns on AutoCalc if it was turned off by mistake.

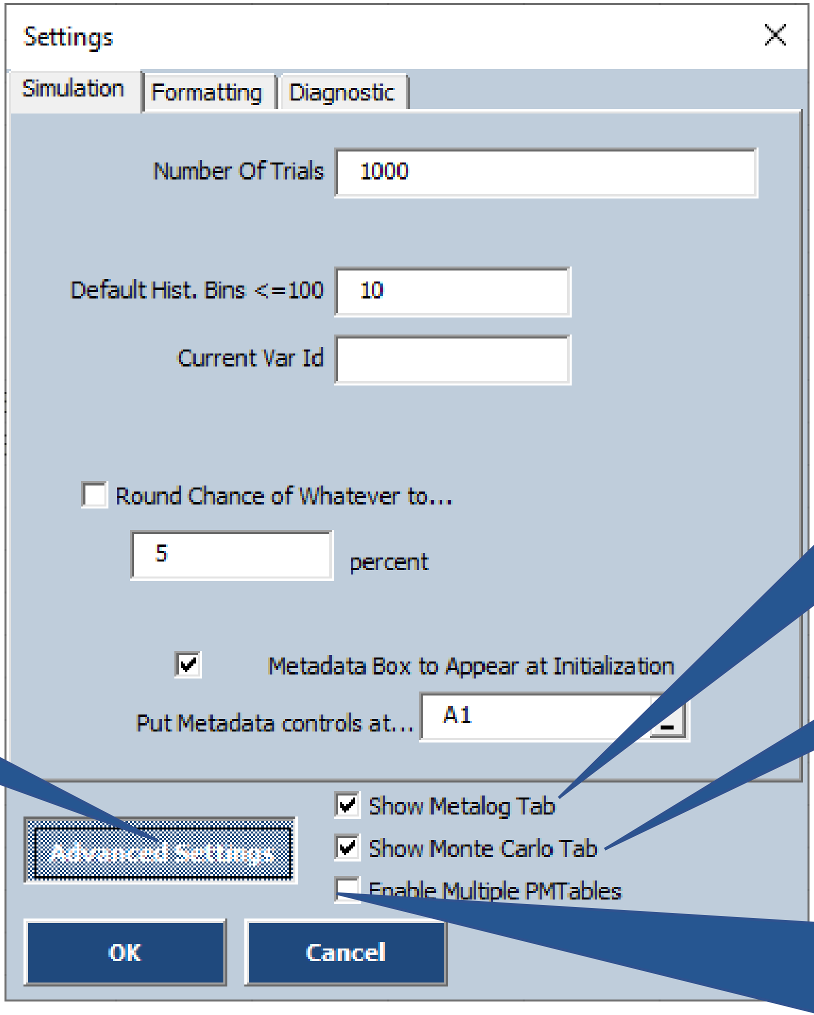
Offers advice when creating SPT random cells

Warn you if AutoCalc is being turned on

Allows creating and switching between multiple PMTable sheets with differing names.



f) Settings – Advanced Settings



Press the Advanced Settings button to show the Advanced Settings options.

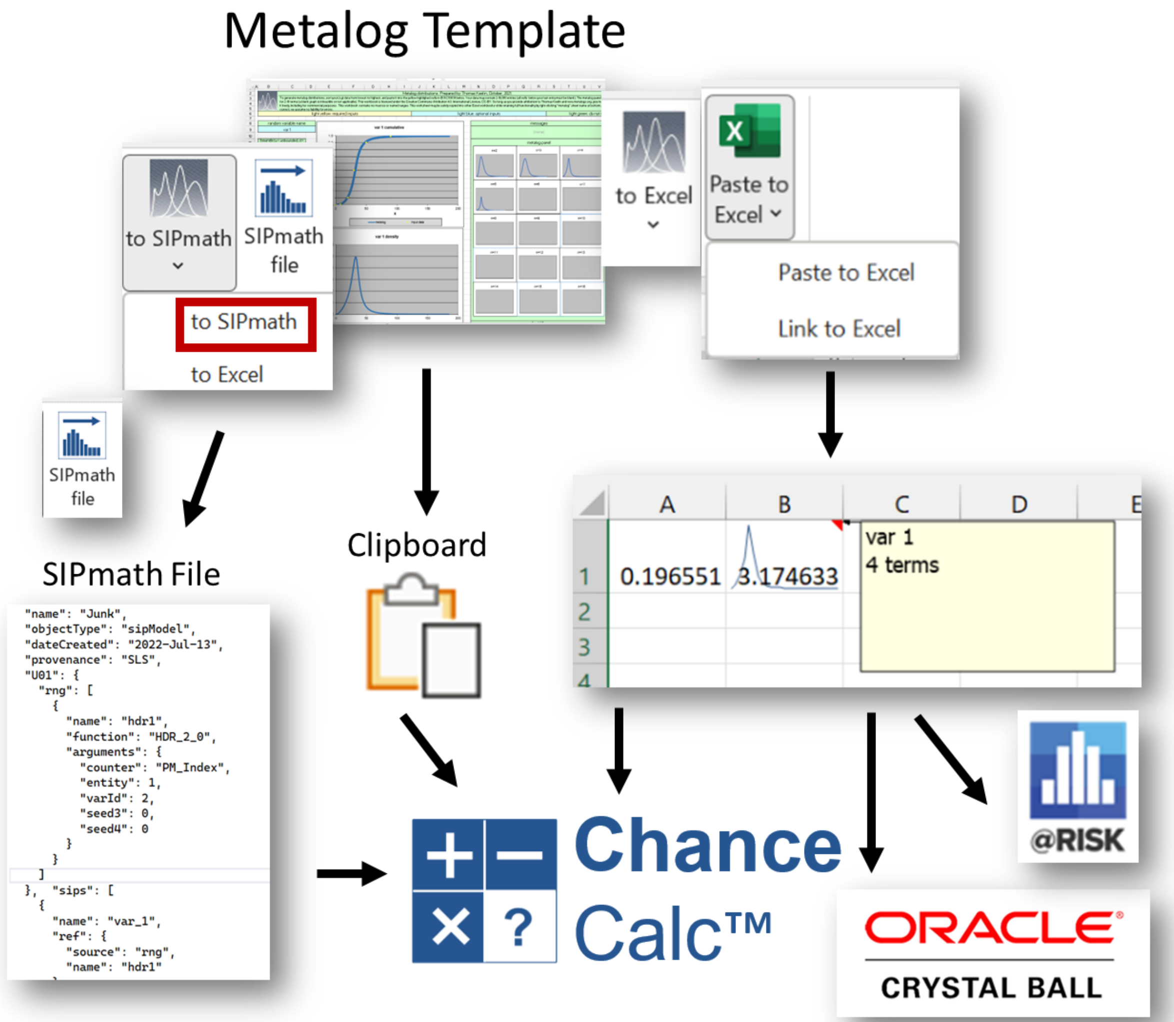
Check the Show Metalog Tab checkbox to show the Metalog section of the Ribbon

Check the Show Monte Carlo Tab checkbox to show the Monte Carlo section of the Ribbon

Check the Enable Multiple PMTables checkbox to enable multiple PMTable pages in your model.

Not recommended for beginners.

3. Metalog Ribbon



3. Metalog Interface

SIPmath and Excel Modes

The interface has two modes with two sub modes as shown in the table below.

Modes Summary

SIPmath Mode		Excel Mode	
to SIPmath to Clipboard to SIPmath to Excel to Clipboard to File	Creates 3.0 Standard JSON stochastic libraries, either copied to the clipboard or as JSON files.	to Excel Paste to Excel to SIPmath to Excel Paste to Excel Link to Excel	Creates Metalog formulas to be either pasted or linked directly to your spreadsheet model.
to Clipboard	Click to specify Library Name, Provenance and HDR Seeds. In your Excel model from the **Chance**Calc menu, use Clipboard option from the SIP Input Command.	Paste to Excel	In your Excel model, click to specify location of Random cell, HDR seeds and optional graph and density data. Metalog formula will be pasted into the model.
to File	Click to specify Library Name, Provenance and HDR Seeds. You will be prompted to save the JSON SIPmath file to your desktop.	Link to Excel	In your Excel model, click to specify location of Random cell, HDR seeds and optional graph and density data. Also specify range for cells linked to parameters in the Metalog template.

3. Metalog Interface

1. Load a metalog template file

2. Click the Metalog tab

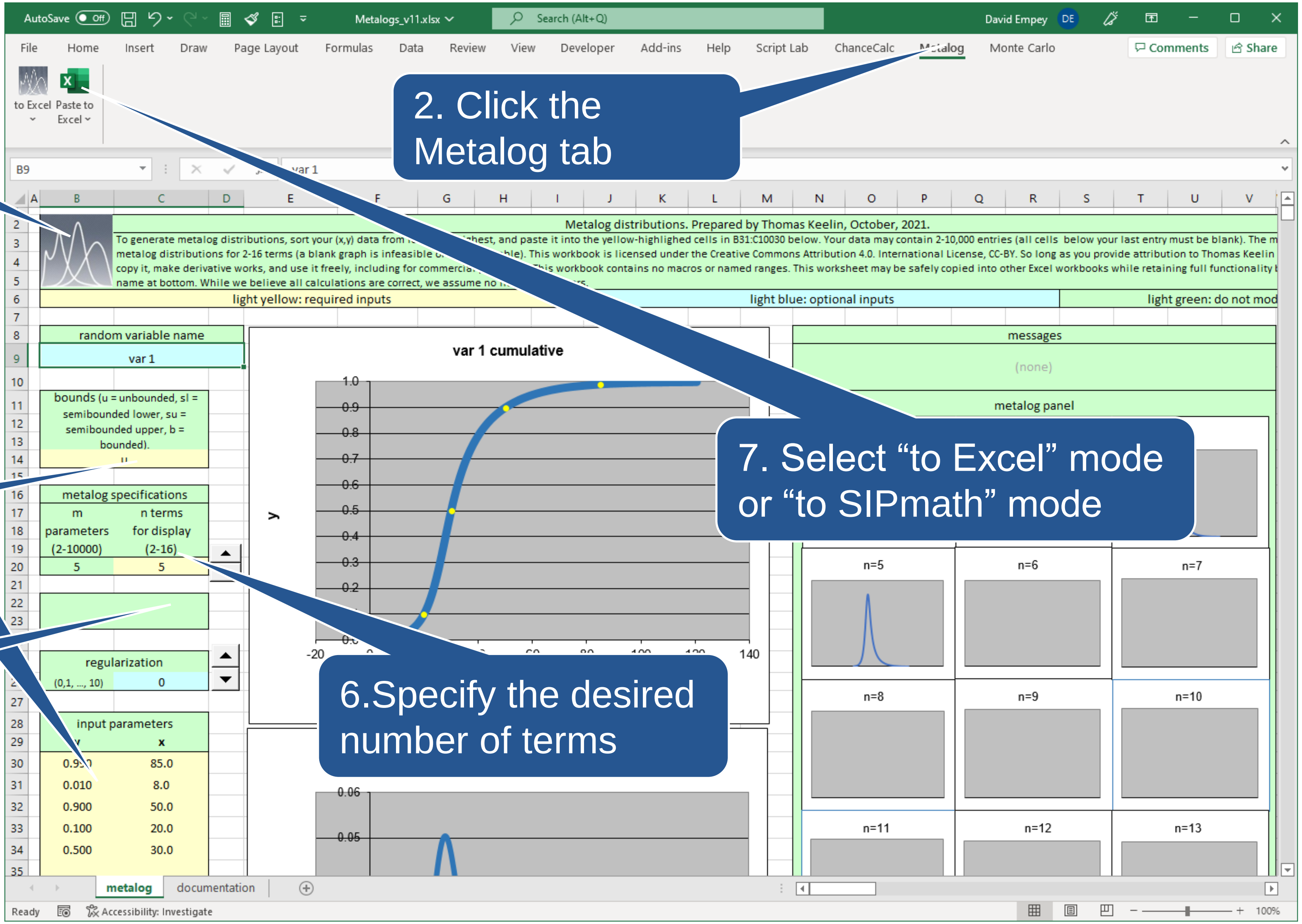
3. Enter the data for your distribution

4. Select the desired boundary type

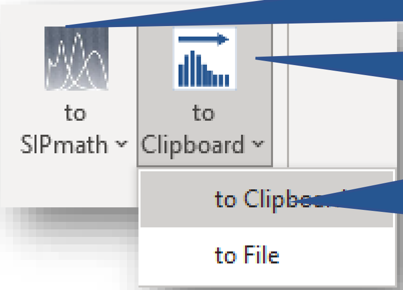
5. Select the desired boundary values

6. Specify the desired number of terms

7. Select “to Excel” mode or “to SIPmath” mode



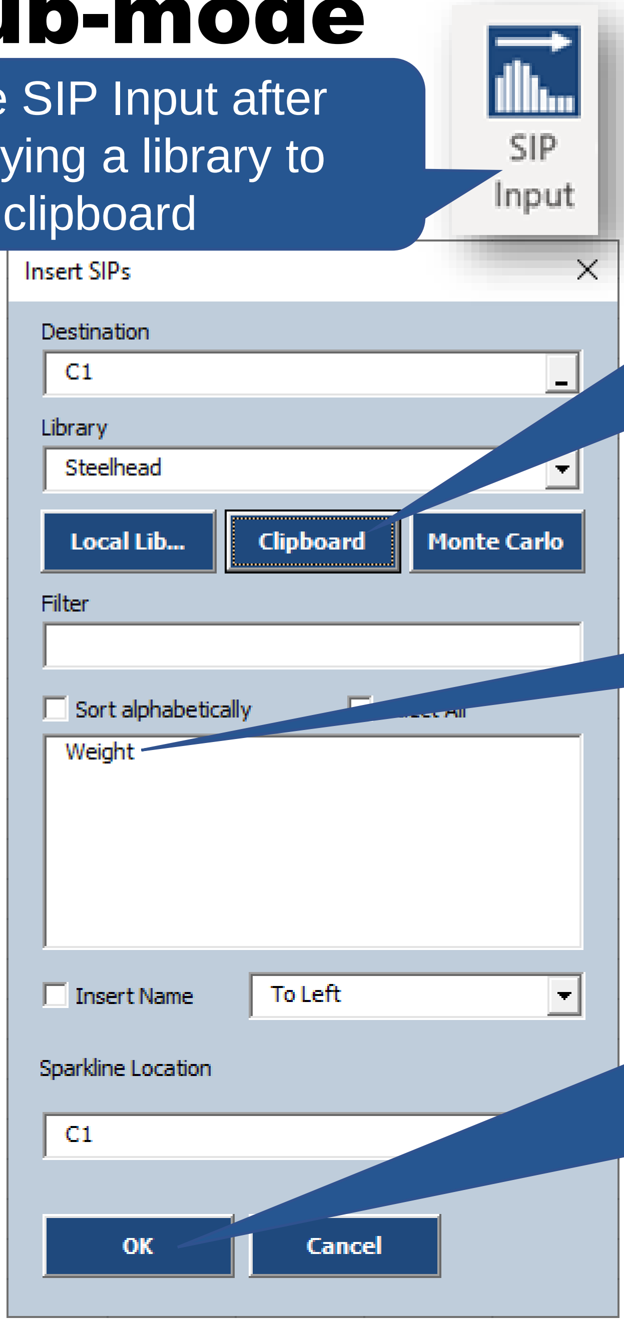
SIPmath mode, Clipboard sub-mode



Button images and text will change depending on the last mode and sub-mode chosen

Create a single-SIP library and copy it to the clipboard for use in the SIP Input dialog

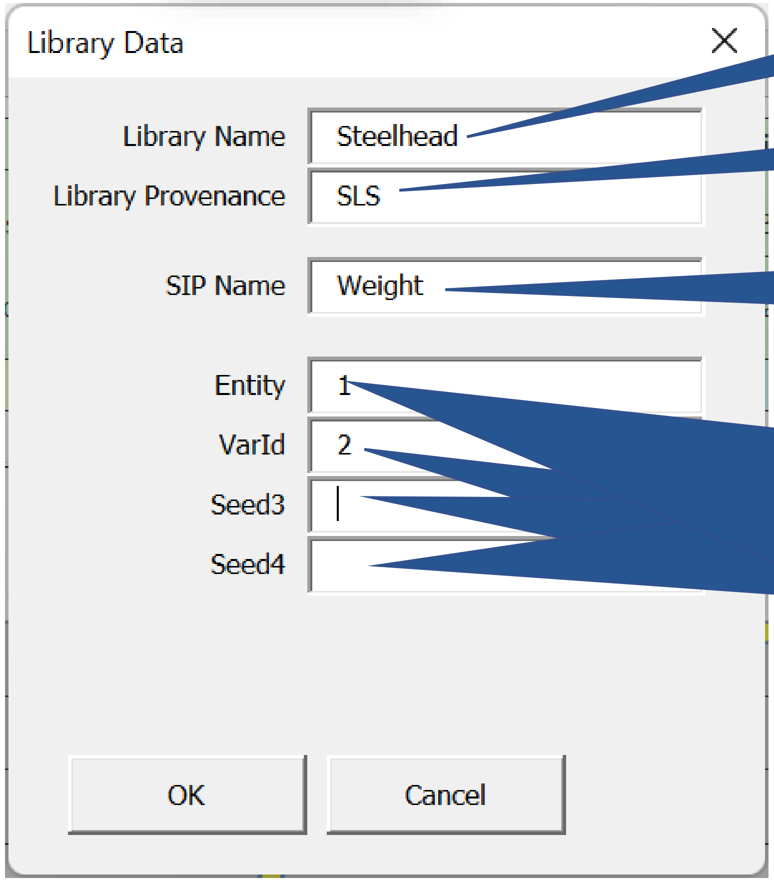
Use SIP Input after copying a library to the clipboard



Press Clipboard to fill dialog with clipboard library data

Make sure to select a SIP

Press OK to insert the Metalog library formula in Excel



Name of the library

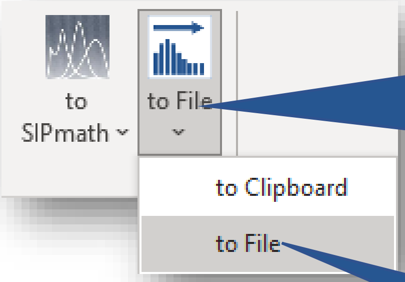
Source of the data

Name of the SIP within the library

Seeds for the HDR random formula used by the Metalog formula to create the library data values.

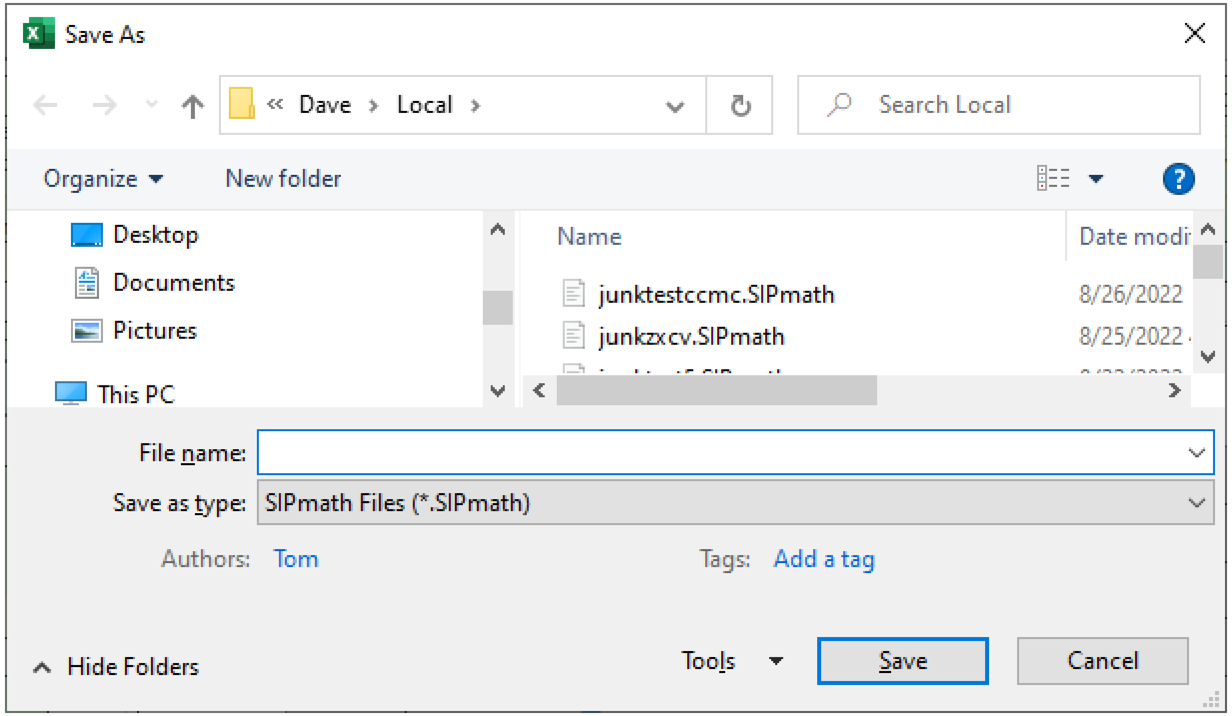
Enter text without quotes in Seed3 or Seed4 for user specified seed.

SIPmath mode, to File sub-mode



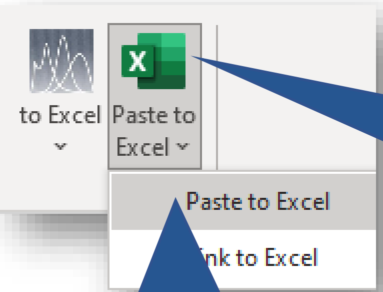
Button images and text will change depending on the last mode and sub-mode chosen

Create a single-SIP library and save it as a .SIPmath 3.0 JSON format file.



This is identical to the Clipboard sub-mode except the following dialog appears to save the file.

Excel mode, Paste sub-mode

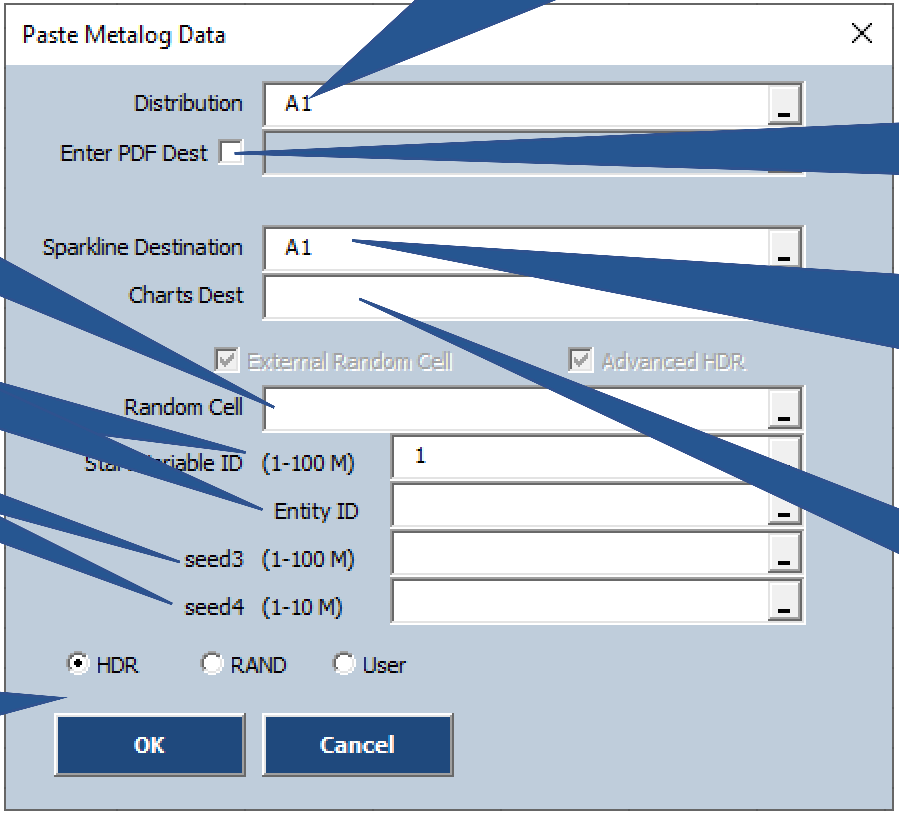


Button images and text will change depending on the last mode and sub-mode chosen

Paste Metalog formulas into Excel without linking to Metalog template

Random cell to be used by Metalog distributions formula

Destination of Metalog distribution formula (you can select this before pressing Paste)



Check box and fill in destination for a PDF formula as well

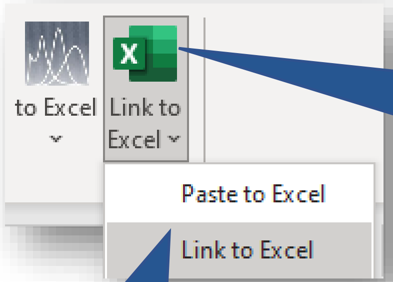
Seeds for use by HDR random cell formula

Destination of sparkline for the distribution. The data for the sparkline will be copied to a tab named Sparkline Data.

Select random cell formula: HDR, Excel's RAND(), or a user formula (Last option leaves the formula in cell untouched)

Location of static CDF and PDF chart images

Excel mode, Link sub-mode



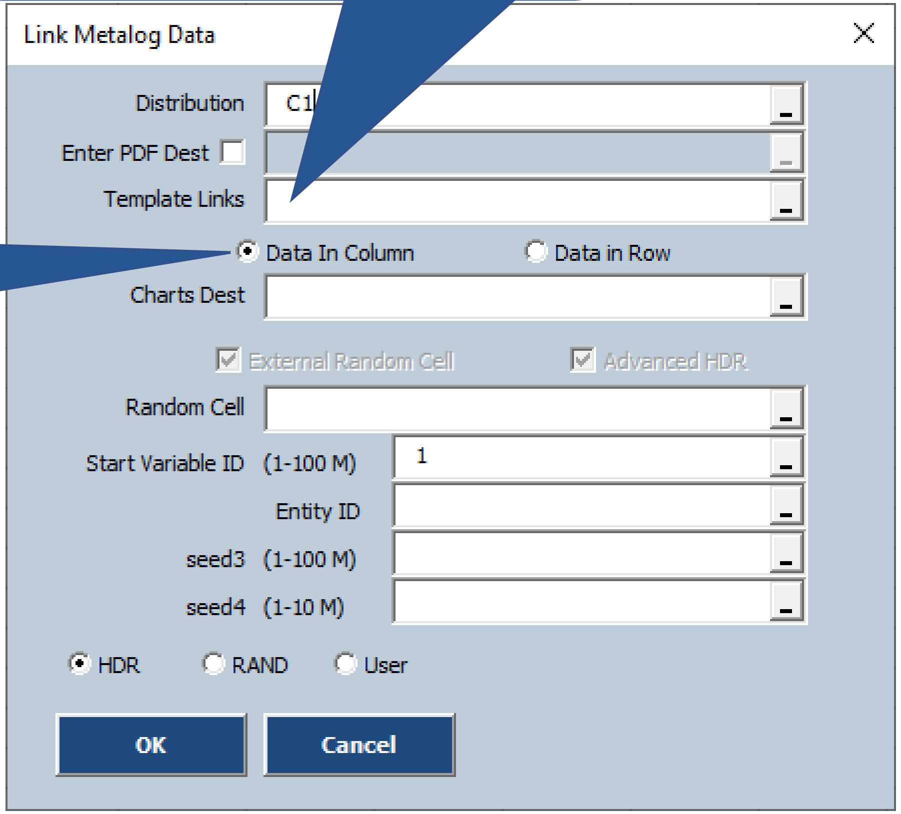
Button images and text will change depending on the last mode and sub-mode chosen

Links from the Metalog template sheet will be copied to this range, including links to sip name, bounds, and a-coefficients.

NOTE: sparklines not directly available in this mode.

Add Metalog formulas into Excel linked to the Metalog template

Select whether linked data will be in a row or a column.



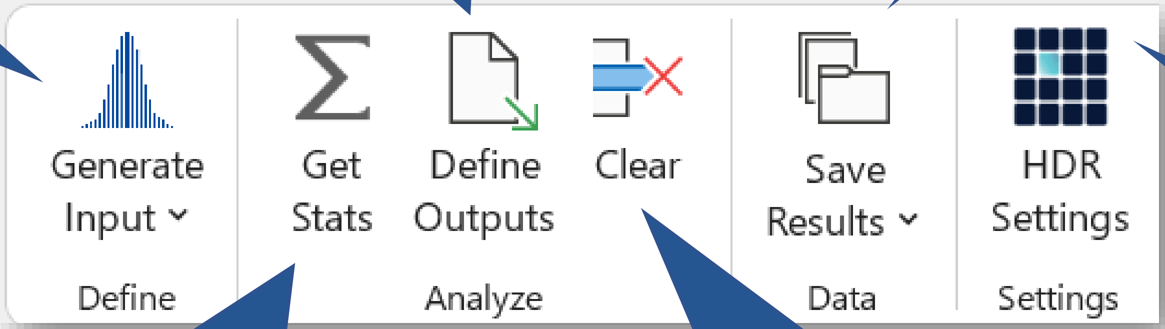
This is otherwise identical to the Paste sub-mode except the metalog formulas and charts are all linked to the Metalog template sheet, and will reflect any changes to that template.

4. Monte Carlo Ribbon

a. Select from a list formulas for any of several standard or custom distributions. (NOTE: this will also initialize the model if necessary.)

c. Defines selected cell as an output. (NOTE: this is usually done automatically when using the ChanceCalc tab. If necessary the model will also be automatically initialized.)

e. Saves current model data in SIPmath 2.0 XLS format, CSV format, or repeatedly saves data while changing a simulation parameter value.



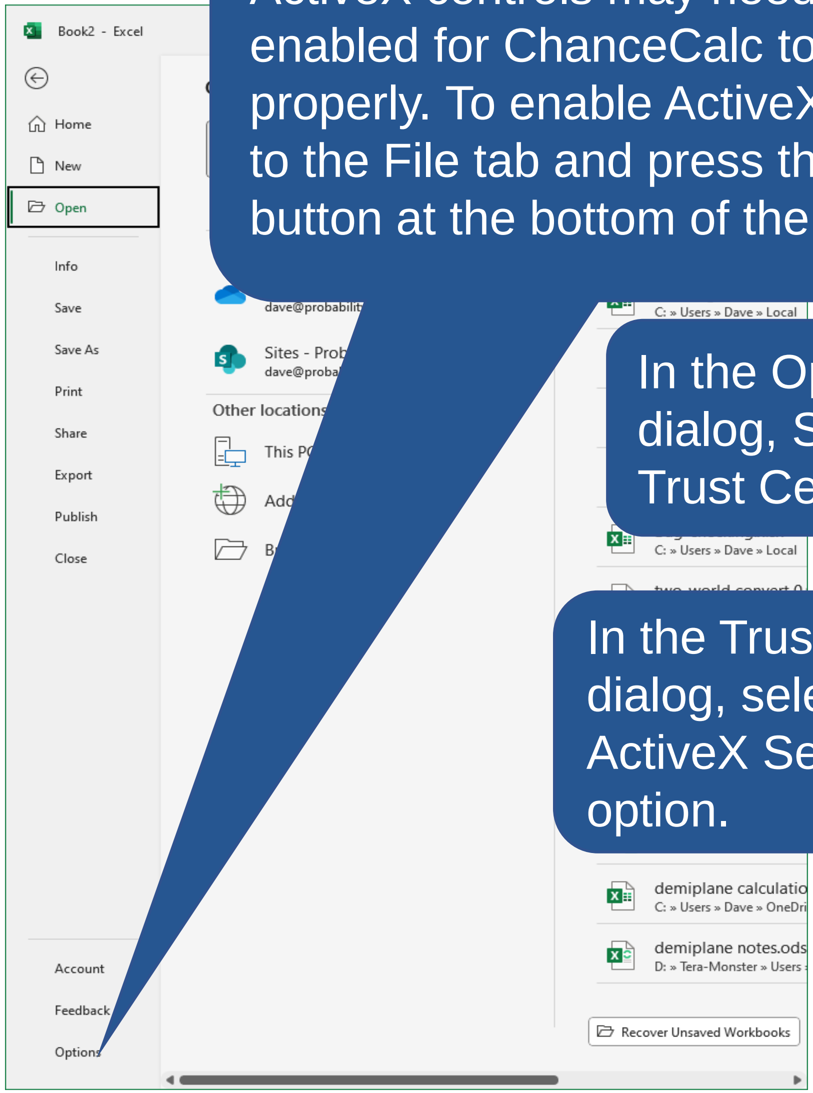
f. HDR Seed Attribute Names and Starting Var ID

b. Converts statistics formulas referring to output cells to refer to the corresponding output SIP range. (NOTE: this DOES NOT automatically initialize outputs.)

d. Un-defines the selected output and removes associated SIP data on PMTable sheet.

Appendix: Enabling ActiveX Controls

ActiveX controls may need to be enabled for ChanceCalc to work properly. To enable ActiveX controls, go to the File tab and press the Options button at the bottom of the screen.



In the Options dialog, Select the Trust Center option.

In the Trust Center dialog, select the ActiveX Settings option.

Select either the third (shown here) or fourth ActiveX Settings option.